

Supplementary Material I

Dataset Description and Synthesis Analysis

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This document provides supplementary details for the main manuscript. Section 1 describes the dental caries dataset, including preprocessing, class definitions, distribution analysis, and structural statistics. Section 2 presents additional synthesis evaluation results, including epoch selection rationale, baseline comparisons, and class-wise generation analysis.

1 Dataset Description and Analysis

1.1 Source and Preprocessing

The dataset used in this work originates from the DC1000 corpus released in the MLUA benchmark [1], comprising 600 full-size panoramic radiograph (PR) images with pixel-level caries annotations. Each PR was processed into non-overlapping Regions of Interest (RoIs) using a 384×384 sliding window, retaining only RoIs that contain at least one caries pixel. This procedure yields **2,615 image-mask pairs** in total, derived from **591 unique source radiographs** (491 for training, 100 for testing). The final data split allocates 1,833 pairs for training, 521 for validation, and 261 for testing.

1.2 Class Definitions and Pixel Encoding

The segmentation masks encode four semantic classes using discrete pixel intensity values, as defined in Table 1:

Table 1: Class definitions and pixel encoding scheme for the dental caries segmentation masks.

Class	Pixel Value	Description
Background	0	Healthy tissue, enamel, bone, and non-carious regions
Superficial Caries (SC)	102	Early-stage demineralization limited to the enamel surface
Medium Caries (MC)	153	Moderate lesion extending into the dentin layer
Deep Caries (DC)	255	Advanced carious involvement reaching near the pulp

1.3 Pixel-Level Class Distribution

The dataset exhibits severe class imbalance, which is characteristic of medical imaging tasks where pathological regions occupy a small fraction of the total image area. Table 2 summarizes the pixel-level statistics across all 2,615 RoIs.

Table 2: Pixel-level class distribution across 2,615 RoI image-mask pairs. Total pixels analyzed: 385,597,440.

Class	Pixel Count	% of Total	Present in (% images)
Background (0)	382,684,349	99.2445%	2,615 (100.00%)
Superficial Caries (102)	359,428	0.0932%	1,575 (60.23%)
Medium Caries (153)	605,006	0.1569%	1,132 (43.29%)
Deep Caries (255)	1,948,657	0.5054%	1,059 (40.50%)

Several observations are noteworthy:

- **Extreme foreground scarcity:** All three caries classes combined account for only 0.7555% of all pixels, with background dominating at 99.24%.
- **Inverse prevalence–area relationship:** SC is the most *common* class (present in 60.23% of images) but occupies the *least* pixel area (0.09%), reflecting the nature of superficial caries as small, early-stage lesions. Conversely, DC is present in only 40.50% of images but contributes the largest pixel area (0.51%), as deep carious lesions tend to be larger.
- **MC occupies an intermediate position:** Medium caries appears in 43.29% of images with 0.16% pixel coverage, consistent with its clinical characterization between early and advanced stages.

Figure 1 visualizes the class presence statistics across the dataset.

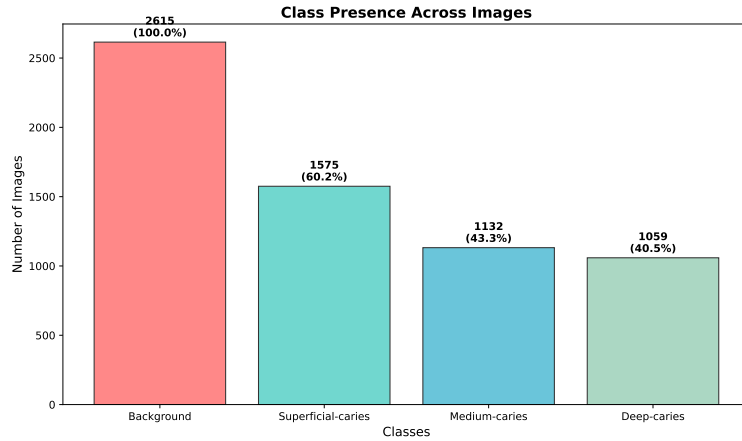


Figure 1: Class presence analysis showing the fraction of images containing each caries class. SC appears in more images (60.23%) than DC (40.50%), despite DC occupying more pixel area overall — highlighting the inverse prevalence–area relationship characteristic of this dataset.

**Class Distribution Among Caries Types
(Percentage of Total Caries Pixels & Component Counts)**

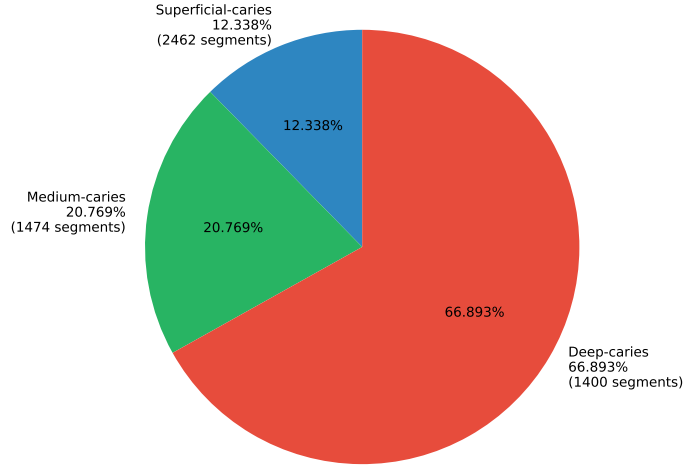


Figure 2: Distribution of images across caries class combination categories. Single-class images dominate the corpus, with SC-only comprising 28.15% of images, followed by DC-only (17.74%) and MC-only (16.29%).

1.4 Class Combination Analysis

Since a given RoI may contain one, two, or all three types of caries simultaneously, we analyze the co-occurrence patterns across the dataset, as visualized in Figure 2. Table 3 presents the full breakdown.

Table 3: Class combination analysis across 2,615 RoI patches. *Image %* denotes the fraction of images in which that combination occurs; *Pixel Area %* denotes that combination’s share of all caries pixels (total: 2,913,091).

Class Combination	Count	Image %	Pixel Area	Pixel %
SC only (102)	736	28.15%	151,318	5.19%
MC only (153)	426	16.29%	206,415	7.09%
DC only (255)	464	17.74%	894,153	30.69%
SC + MC (102 + 153)	394	15.07%	317,228	10.89%
MC + DC (153 + 255)	150	5.74%	355,313	12.20%
SC + DC (102 + 255)	283	10.82%	571,924	19.63%
All three (102 + 153 + 255)	162	6.20%	416,740	14.31%

Key observations from the class combination analysis:

- **Single-class dominance:** 62.18% of images (1,626/2,615) contain only one caries type. Among these, SC-only is the most frequent (28.15%), followed by DC-only (17.74%) and MC-only (16.29%).
- **DC dominates pixel area:** Despite appearing as the sole class in only 17.74% of images, DC-only accounts for 30.69% of all caries pixels — the largest single contribution. This reflects the typically larger spatial extent of deep carious lesions.
- **Multi-class images are less common:** Two-class combinations constitute 31.63% of the dataset (SC+MC: 15.07%, SC+DC: 10.82%, MC+DC: 5.74%), while only 6.20% contain all three types simultaneously.

- **Rarest combination:** MC+DC appears in only 5.74% of images, making it the least frequent two-class combination, yet contributes 12.20% of caries pixels.

Figure 3 presents the class combination analysis in tabular and visual form.

Class Combination Analysis
Distribution of Caries Type Combinations Across Images with Pixel Areas

Class Combination	Image Count	Image %	Total Pixel Area	Pixel Area %
Background only (no caries)	0	0.00%	0	N/A (background)
Superficial-caries only (102)	736	28.15%	151,318	5.19%
Medium-caries only (153)	426	16.29%	206,415	7.09%
Deep-caries only (255)	464	17.74%	894,153	30.69%
Superficial + Medium (102 + 153)	394	15.07%	317,228	10.89%
Medium + Deep (153 + 255)	150	5.74%	355,313	12.20%
Superficial + Deep (102 + 255)	283	10.82%	571,924	19.63%
All three types (102 + 153 + 255)	162	6.20%	416,740	14.31%
Other/unexpected combinations	0	0.00%	0	0.00%

Figure 3: Class combination analysis showing the distribution of single-class and multi-class RoI patches. The majority of images contain a single caries type, while images with all three types (SC+MC+DC) are the rarest.

1.5 Class Imbalance Analysis

The imbalance across the three caries classes has direct implications for both synthesis and segmentation. Figure 4 presents a comprehensive visualization of how the imbalance manifests at both the image and pixel levels.

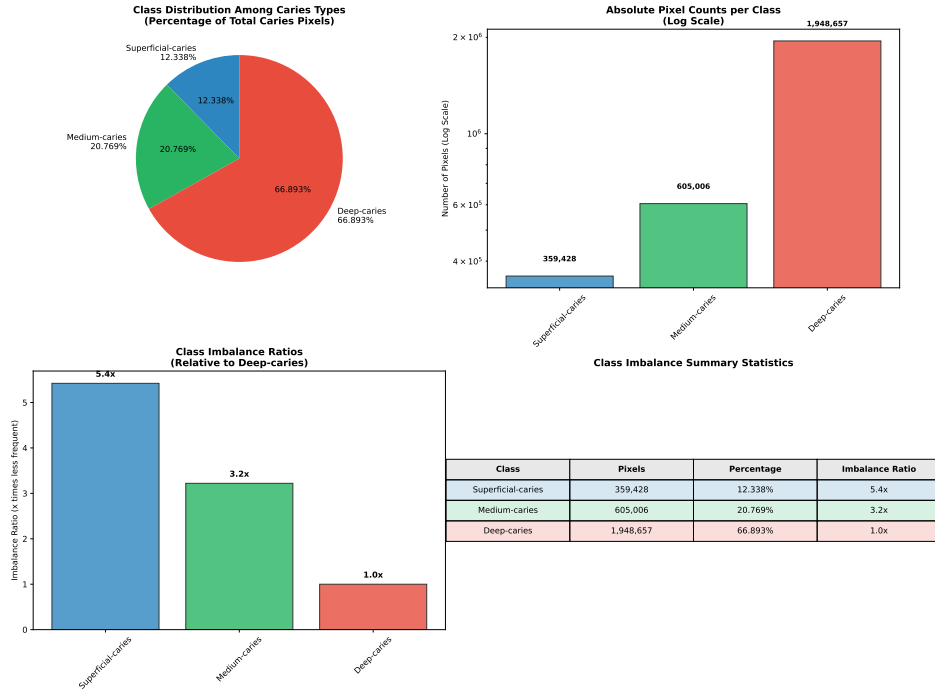


Figure 4: Class imbalance analysis across the dataset. The visualization highlights the disparity between class occurrence frequency (image-level) and spatial coverage (pixel-level), where SC is prevalent but small, and DC is less frequent but spatially dominant.

1.6 Connected Components Analysis

To characterize the structural complexity of caries annotations, we analyze the number of connected components per class per image. This provides insights into lesion fragmentation and spatial distribution.

bution. Table 4 summarizes the statistics, and Figure 5 visualizes the full distribution.

Table 4: Connected components statistics per caries class. *Avg. per image* is computed over all 2,615 images (including those without the class, which contribute 0 components).

Class	Total Components	Avg. per Image	Max	Images with ≥ 1
SC (102)	2,462	0.94	9	1,575 (60.23%)
MC (153)	1,474	0.56	7	1,132 (43.29%)
DC (255)	1,400	0.54	13	1,059 (40.50%)

Several structural patterns emerge:

- **SC has the highest fragmentation:** With an average of 0.94 components per image and a maximum of 9, superficial caries tends to appear as multiple small, scattered lesions. Among SC-containing images, 23.05% have two or more components.
- **MC and DC are more spatially consolidated:** Both MC and DC average approximately 0.55 components per image. When present, they typically appear as a single contiguous region (MC: 76.06% single-component when present; DC: 78.09%).
- **Rare highly fragmented cases:** A few images contain exceptionally fragmented DC annotations (up to 13 components), likely representing multi-focal deep carious involvement across several teeth within a single RoI.

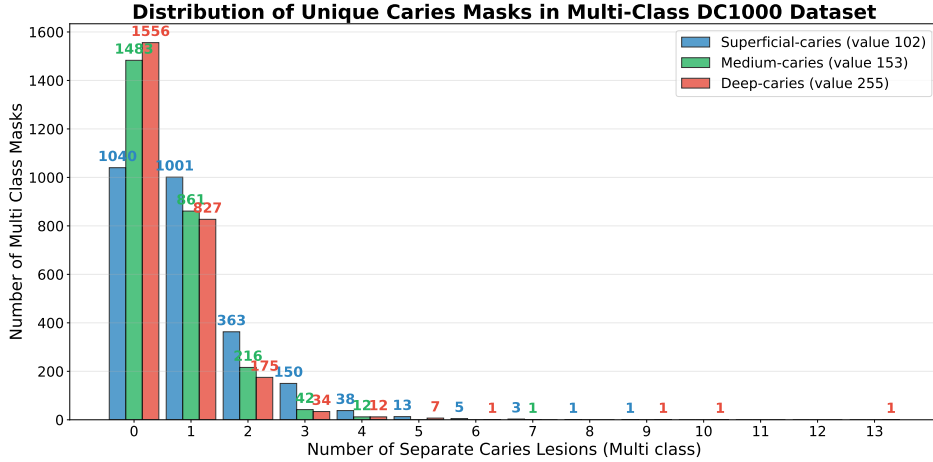


Figure 5: Distribution of connected components per class across all 2,615 images. SC shows the widest spread, reflecting its tendency to appear as multiple small lesions, while MC and DC are predominantly single-component when present.

1.7 Dataset Visualization

Figure 6 presents a mosaic of representative RoI samples from the dataset, illustrating the diversity of caries manifestations across images.

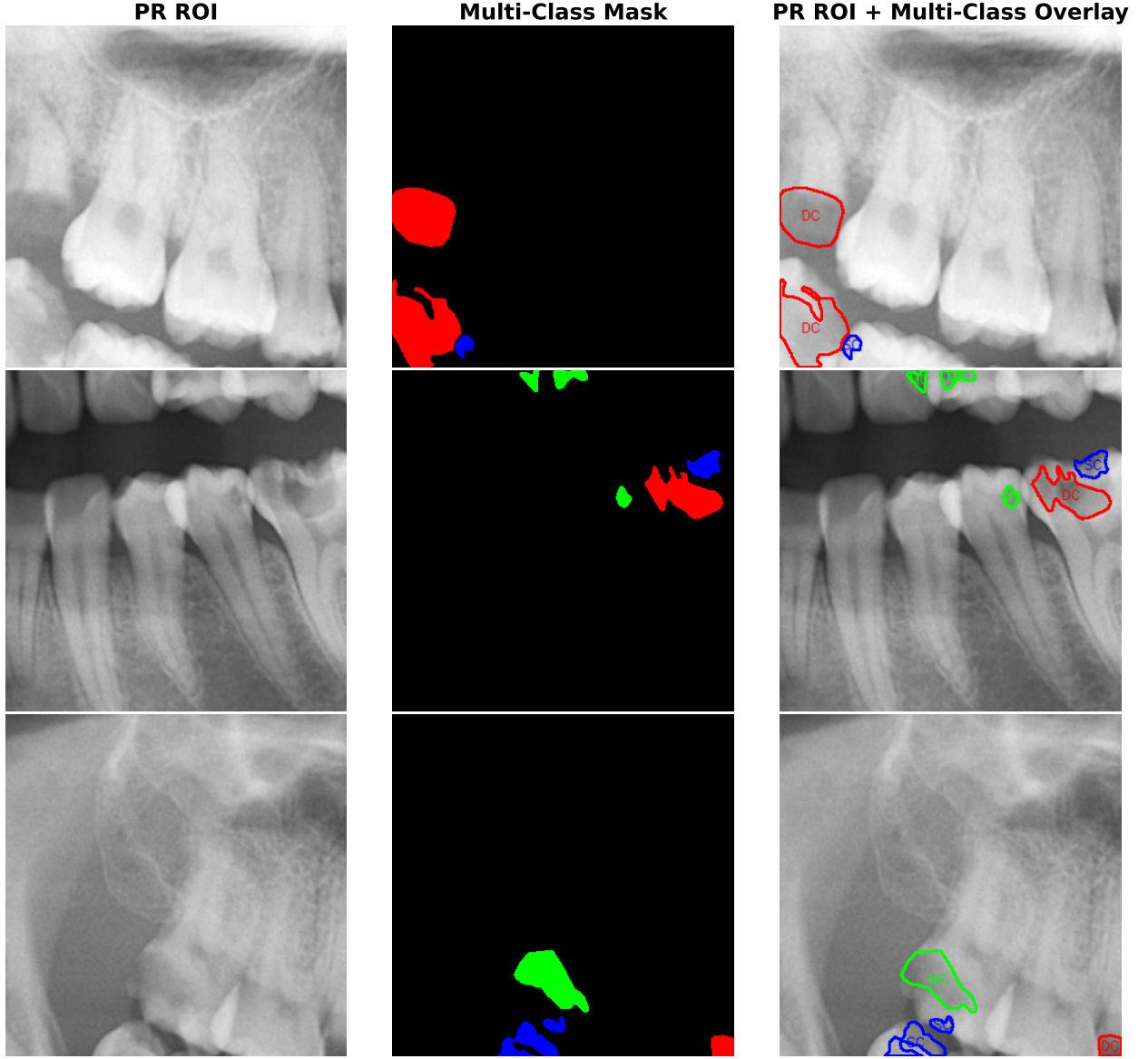


Figure 6: Mosaic of representative RoI samples from the dental caries dataset. Images are overlaid with color-coded class boundaries: **Blue** for superficial caries (SC), **Green** for medium caries (MC), and **Red** for deep caries (DC). The samples showcase the diversity in lesion size, shape, severity, and spatial arrangement across the dataset.

2 Synthesis Evaluation

2.1 Training Dynamics and Epoch Selection

The *CarDiff* model was trained for 1,000 epochs, with evaluation metrics computed at regular intervals across training, validation, and test splits. Figure 7 presents the epoch-wise progression of four evaluation metrics: Fréchet Inception Distance (FID), Kernel Inception Distance (KID), Inception Score (IS), and Learned Perceptual Image Patch Similarity (LPIPS).

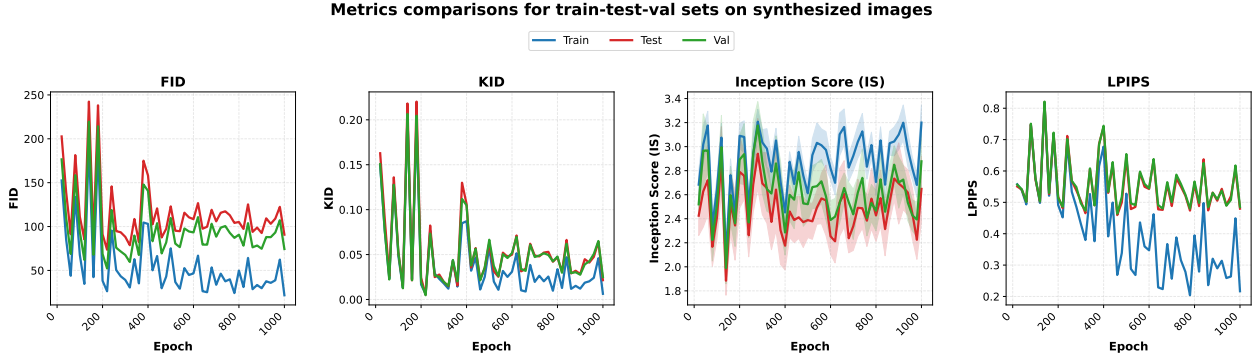


Figure 7: Epoch-wise *CarDiff* metrics — FID ($\downarrow$), KID ($\downarrow$), IS ($\uparrow$), and LPIPS ($\downarrow$) — evaluated on training (blue), testing (red), and validation (green) synthesized radiographs. Epoch 220 is identified as the optimal checkpoint, balancing synthesis quality across all splits.

Based on the comprehensive evaluation across all three splits, **epoch 220** was selected as the best-performing checkpoint. It achieves the lowest FID and KID on both test and validation sets, indicating excellent generalization and distribution alignment with real data. While some later epochs (e.g., 780, 1000) show better training LPIPS, their test and validation metrics are significantly worse, suggesting overfitting. Epoch 220 strikes the optimal balance across all four metrics and is the most robust choice for downstream applications.

2.2 Baseline Comparison

Table 5 presents the quantitative comparison of *CarDiff* against four baseline synthesis methods. All models were evaluated by generating radiographs conditioned on the 261 test masks using 1,000 denoising steps for diffusion-based methods.

Table 5: Quantitative comparison of synthesis methods. FID ($\downarrow$), KID ($\downarrow$), and LPIPS ($\downarrow$): lower is better. IS ($\uparrow$): higher is better. **Bold**: best value per metric.

Method	FID $\downarrow$	KID $\downarrow$	IS $\uparrow$	LPIPS $\downarrow$
CarDiff (ours)	90.76	0.0215	2.65	0.480
SCDM	95.55	0.0165	2.75	0.514
SISDM	99.96	0.0241	2.40	0.513
SegDiff	126.72	0.0713	2.21	0.636
SPADE	164.05	0.0915	2.18	0.517

The results demonstrate that *CarDiff* achieves the best FID (90.76) and LPIPS (0.480) among all baselines, representing a 5.0% and 6.6% improvement over the next-best method (SCDM), respectively. SCDM achieves marginally better KID (0.0165 vs. 0.0215) and IS (2.75 vs. 2.65), making it a competitive runner-up. SISDM performs reasonably across all metrics. SegDiff and SPADE significantly lag behind on FID and KID, with SPADE showing the weakest overall performance (FID: 164.05, KID: 0.0915). The clear hierarchy — *CarDiff* > SCDM > SISDM $\gg$ SegDiff > SPADE — validates the effectiveness of the proposed causally-structured diffusion approach.

2.3 Qualitative Synthesis Comparison

Figure 8 presents a side-by-side visual comparison of synthesized radiographs across all five methods, conditioned on the same input label masks. Each column shows the output of a different model for the same set of test inputs, with class-coded boundary overlays: Blue (SC), Green (MC), and Red (DC).

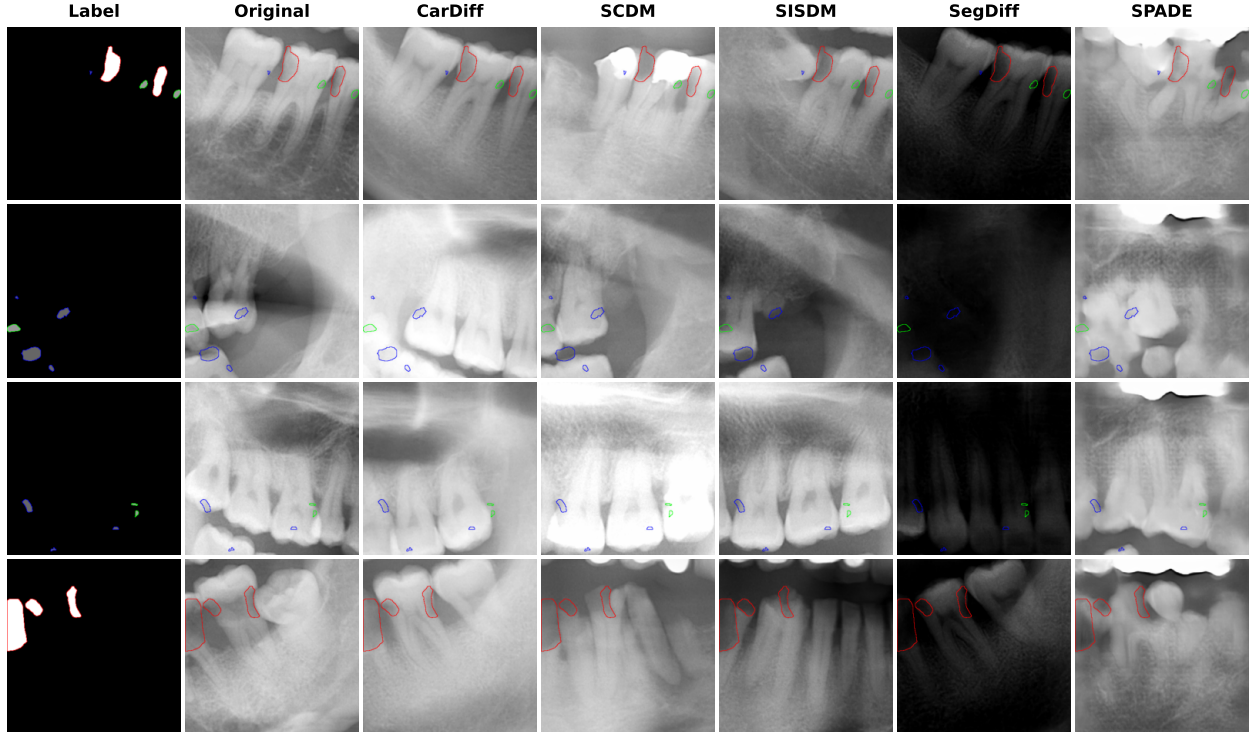


Figure 8: Qualitative comparison of synthesized dental radiographs across all models. Columns from left to right: Label mask, Original image, CarDiff (ours), SCDM, SISDM, SegDiff, and SPADE. Class boundaries are overlaid in color: Blue for SC, Green for MC, and Red for DC. *CarDiff* produces radiographs with more realistic attenuation gradients and sharper caries boundaries compared to other methods.

2.4 Class-Wise Generation Analysis

Beyond aggregate metrics, we evaluated synthesis quality for individual caries classes and their combinations. This analysis is crucial for understanding whether the model can faithfully generate all three caries severity levels, particularly the under-represented SC and MC classes. Figure 9 presents the combined class-wise metrics across all evaluation criteria.

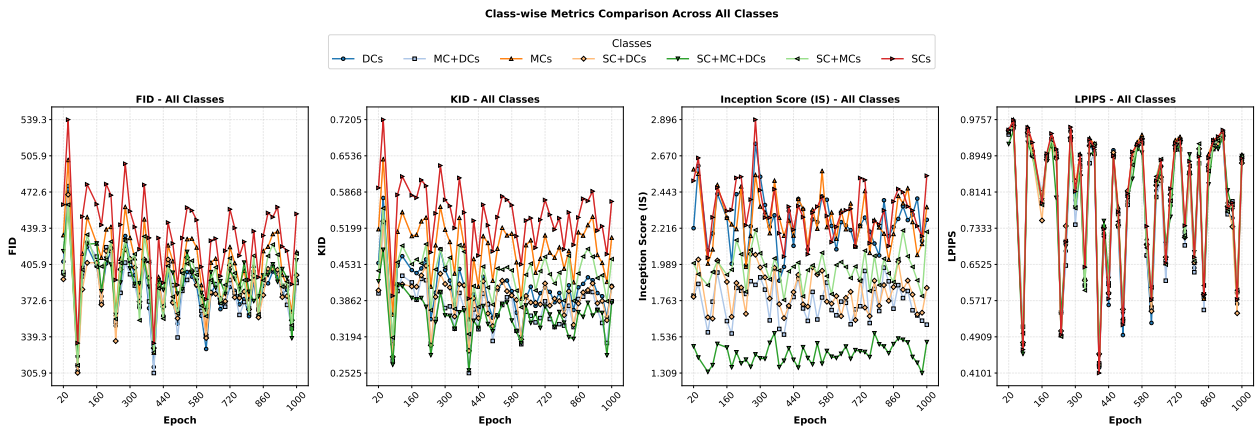


Figure 9: Class-wise synthesis evaluation metrics (FID, KID, IS, LPIPS) across individual caries classes (SC, MC, DC) and their combinations (SC+MC, SC+DC, MC+DC, SC+MC+DC). The results demonstrate that individual class generation is particularly challenging for DC, while using class combinations during training significantly improves generation quality across all metrics.

The class-wise analysis reveals several important patterns:

- **Individual class generation is challenging:** Generating radiographs conditioned on single-class masks (SC-only, MC-only, or DC-only) yields higher FID and worse perceptual quality compared to multi-class combinations. This is expected, as single-class masks are less representative of natural radiographic anatomy.
- **DC is the most difficult to generate in isolation:** Despite occupying the largest pixel area, DC-only generation produces the highest FID among single-class conditions. This likely reflects the complex intensity gradients associated with deep caries penetrating the dentin.
- **Multi-class combinations improve quality:** Combining classes (e.g., SC+DC, SC+MC+DC) during conditioning substantially improves all metrics, suggesting that the model benefits from richer spatial context when multiple caries types co-occur.
- **SC+MC+DC provides the best context:** The three-class combination achieves the most stable and favorable metrics across all criteria, supporting the use of diverse, multi-class masks for synthesis.

2.5 Synthesis for Data Augmentation

Using the best model checkpoint (epoch 220), *CarDiff* was employed to synthesize 13,075 radiographs conditioned on procedurally generated segmentation masks. The mask generation process involved: (1) extracting lesion patches (connected components) from real training masks, (2) recombining them into new synthetic masks by placing them randomly within a 384×384 canvas with a random count between one and five components, and (3) using *CarDiff* to generate corresponding radiographs. This produces a diverse pool of unique mask-image pairs suitable for augmenting the training set.

The augmentation study, presented in detail in Supplementary Material II, evaluates five augmentation levels ($1\times$ through $5\times$ the original training data) across seven segmentation architectures (U-Net, U-Net++, FPN, LinkNet, MA-Net, PAN, PSPNet), all with ResNet-34 encoders pretrained on ImageNet. The results demonstrate consistent improvements, with the $5\times$ augmentation achieving a 6.5% average Dice score improvement across all models, with class-wise gains of 25.7% for MC, 1.6% for SC, and 16.6% for MC+SC — validating synthetic data as an effective strategy for addressing class imbalance in dental caries segmentation.

References

- [1] Wang, Y., et al.: Multi-label dental caries segmentation from panoramic radiographs. In: Medical Image Analysis (2023)